

Assignment: Human Pose Detection with YOLO Model

Objective:

The goal of this assignment is to evaluate the candidate's ability to apply the Ultralytics YOLO model to detect human poses from an input image and specifically identify and extract details related to human hands.

Instructions:

1. Input:

- A single image will be provided containing one or more humans.

2. Task:

- Use the latest Ultralytics YOLO model to detect human poses.
- Identify keypoints corresponding to human hands (both left and right hands).
- Highlight these keypoints visually on the image.
- Output the coordinates of the detected hand keypoints in a JSON format.

3. Requirements:

- Install and use the Ultralytics YOLO library.
- Use pre-trained weights for pose detection.
- Write clean, modular code with proper comments.

4. Deliverables:

- **Code:** Provide a Python script or Jupyter Notebook with the implemented solution.
- **Annotated Image:** The input image with human hand keypoints clearly marked.
- **JSON Output:** A file containing the coordinates of detected hands in the following format:

```
5. {  
6.   "hands": [  
7.     {"imagename": "1.jpg", "hand1": [x1, y1], "hand2": [x2, y2]}  
8.   ]  
9. }
```

10. Additional Instructions:

- Include a README file explaining:

- How to run the code.
- Dependencies and installation steps.
- Any assumptions or limitations of the approach.
- The solution should handle images with multiple people and provide hand keypoints for each detected person.

11. Evaluation Criteria:

- Correctness of the solution (pose detection and hand identification).
- Code quality and readability.
- Clear and concise documentation.
- Efficiency of the solution.

Test Case Output (For reference)

1. Output with yolo pose detection



2. Output with only hands



3. JSON output

```
4. {  
5.   "hands": [  
6.     {"imagenname": "1.jpg", "hand1": [2548, 2649], "hand2": [3639, 1578]}  
7.   ]  
8. }
```